



Ecological risks assessment of sulfur and heavy metals in sediments in a historic mariculture environment, North Yellow Sea

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ABSTRACT

The environment behaviors of sulfur and heavy metals in sediments are closely related to sediment aging in mariculture area. In this study, the distributions and ecological risks of reduced inorganic sulfur (RIS) and heavy metals were investigated, along with the relationships between different occurrences of RIS and heavy metals. The results indicated that the adequate organic matter in mariculture sediments significantly enhanced the accumulation of acid volatile sulfur (AVS) compared to the control area. In shellfish farming area, biological sedimentation contributed to accumulation of AVS. The chromium (II)-reducible sulfur (CRS) was the main component of RIS in mariculture area. The environmental risks of heavy metals in mariculture area presented low levels. Principal component analysis (PCA) showed that distribution of Cu closely related to mariculture activities compared to other heavy metals. For ecological risks of heavy metals, the ratio of $\sum(\text{acid-soluble fraction (F1)} + \text{reducible fraction (F2)} + \text{oxidizable fraction (F3)})/\text{AVS}$ was the appropriate index rather than conventional simultaneous extraction of heavy metals (SEM)/AVS, because SEM/AVS would overestimate the toxicity of heavy metals. AVS/RIS ratios significantly positively correlated with Pb ($F2/(F1 + F2 + F3 + \text{residual fraction (F4)})$), $F2/\sum F$, Pb ($F3/\sum F$), and Zn ($F3/\sum F$), while significantly negatively correlated with Pb ($F4/\sum F$) and Cu ($F1/\sum F$). These results indicated that the accumulation of AVS during the mariculture process was conducive to the formation of F2 and F3 of Pb, and F3 of Zn, conversely to the formation for F4 of Pb and F1 of Cu, because it was opposite to the accumulation of CRS.

1. Introduction

Global mariculture output increase abruptly in recent decades with 20 % of aquatic product for human consumption (FAO, 2020). There is 18,000 km coastal line with a large scale ocean area in China, so the mariculture industry plays an important role in global aquaculture output. For its high health protection function, the sea cucumber has become an important commercial marine aquaculture species in China, and it has been cultivated on a large scale (Hou et al., 2019). Based on the report of Fisheries and Fisheries Administration Bureau of the Ministry of Agriculture and Rural Affairs (2021), by 2020, the mariculture area and yield of sea cucumbers in China reached 24.3×10^4 hm² and 19.66×10^7 kg, respectively. However, with the rapid development of sea cucumber mariculture industry, sediment deterioration, frequent disasters and low efficiency in aquaculture areas are drawing

attention. Sediment plays an important role in the aquaculture ecosystem, the pollutants such as sulfur and heavy metals in sediment are closely related to the productivity of aquaculture area (Yi et al., 2011). In marine sediments, sulfur containing compounds include organic and inorganic forms (Jiang et al., 2021a). For inorganic sulfur, it is mainly active reduced inorganic sulfur (RIS), including acid volatile sulfur (AVS), chromium (II)-reducible sulfur (CRS), and elemental sulfur (ES). AVS is mainly from the dissimilatory sulfate reduction, it is highly reactive and may endanger aquatic organisms through toxicity and oxygen consumption (Kang et al., 2016). CRS is a stable form of RIS, the transformation from AVS to CRS is typically controlled by the organic matter (Morgan et al., 2012), the availability of dissolved sulfate (Jiang et al., 2021a), and reactivity of Fe minerals buried in sediments (Yang et al., 2020). AVS and CRS are the primary fractions of RIS in sediments (Jiang et al., 2021b). ES is the product of incomplete oxidation of RIS, it

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can react with AVS to form CRS (Sheng et al., 2015). Heavy metals are attracting special attention because of their indestructible and persistent properties in the environment, and have high toxic effects on organisms and human health (Fan et al., 2020). The environmental behaviors and biological effectiveness of heavy metals are closely related to their fugitive forms, which are usually classified as acid-soluble fraction (F1), reducible fraction (F2), oxidizable fraction (F3), and residual fraction (F4). These fractions would be transformed with the change of environmental conditions in sediments, resulting in the different ecotoxicities significantly (Chen et al., 2020).

Sulfur in sediments always closely associates with many heavy metals. The transformation of different forms of RIS may influence environmental behaviors and ecotoxicology of heavy metals in different chemical speciation (Chen et al., 2020). AVS may control the geochemical behavior of heavy metals in sediments (Chen et al., 2020), restricting the toxicity (Shyleshchandran et al., 2018), mobility (Jiang et al., 2021b), and bioavailability of heavy metals (Chai et al., 2017), especially for Pb, Cu, Zn, and Cd. Furthermore, the conversion of AVS to CRS may affect the morphological changes of Pb, Cu, Zn, and Cd (Burton et al., 2006), such relationship potentially plays an important role in sediment aging in mariculture areas. Generally, accumulations of AVS and heavy metals in sediments are considered as the typical characteristics of sediment aging (Yokoyama, 2003; Zhang et al., 2012). Previous studies about sulfur and heavy metals in marine sediments mainly focused on the ratio of AVS to simultaneous extraction of heavy metals (SEM), determining the ecological risk of heavy metals (Chen et al., 2020). However, the evaluation results often appear large deviations, because the organic matter in sediment will chelate with heavy metals to form an organic binding state, decreasing the SEM actual value (Wang et al., 2019). Furthermore, when the AVS presents low levels, the SEM/AVS value may distort the evaluation results (Zhang et al., 2019). Therefore, due to the high total organic carbon (TOC) in the sediment, it is difficult to accurately assess the ecological risk of heavy metals by the SEM/AVS ratios in the mariculture area, especially in a historic mariculture region. However, other methods such as $\sum(F1 + F2 + F3)/AVS$ could be an appropriate method to assess the ecological risk of heavy metals in sediments. In addition, different geochemical fractions of sulfur and heavy metals may transform by themselves and make mutual coupling each other, including sulfide bond state formation between sulfur and heavy metals. However, the form changes and coupling mechanism between sulfur and heavy metals during mariculture sediment aging are not well understood. The objectives of this study were to (1) investigate the fraction variations of RIS and heavy metals in

mariculture sediments, (2) assess ecological risks of sulfur and heavy metals, and (3) identify the migration transformation behavior and coupling relationship between sulfur and heavy metals during aging of mariculture sediments.

2. Material and methods

2.1. Study area description and sampling

The study area locates in Sishili Bay and its adjacent area with an average water depth of <15 m in North Yellow Sea (Fig. 1). The water temperature in the Bay is 2.5–3.5 °C in winter and 23.3–27.4 °C in summer, and there is no thermal stratification (Wang et al., 2013). The main tidal components in the Bay are semi-diurnal tidal currents, and the tidal range is 1.66 m. The hydrodynamic process is mainly controlled by wind-induced currents and tidal movement. Superior natural conditions make the bay an important area of aquaculture in north China. The bay has a long history of aquaculture (~100 ys), and aquaculture area once accounted for 70 % of the whole bay. The western part of the bay (S1–S8) is dominated by shellfish farming and the eastern part (S9–S16) by sea cucumber farming. Due to a long period of extensive breeding, the sediments in the aquaculture area are aging, and some areas are even unsuitable for aquaculture.

Sampling was conducted in June 2021, totally 17 surface sediment samples (0–10 cm) in study area were collected using a Box-typed mud collector. To assess ecological risks of sulfur and heavy metals in mariculture area, a site (outside mariculture area) unaffected by any known local breeding activities or anthropogenic impact was selected as a control point (B0). Samples were stored in zip-lock plastic bags on ice in an adiabatic box, and were transported to the laboratory within 8 h. Samples were frozen at −20 °C until further processing.

2.2. Sample analysis

2.2.1. Physic-chemical properties

The pH, electrical conductivity (EC), and salinity in sediments were measured by a supernatant of 1:5 (w/v) suspension extracted with ultrapure water. TOC and total nitrogen (TN) in sediments were measured by an elemental analyzer (Vario MACRO cube, Elementar, Germany). The relative standard deviation (RSD) of the TOC and TN measurements was ≤0.5 %. Before analysis, the samples were freeze-dried and sieved (100 mesh). For TOC determination, sample was acidified with 10 % HCl for 16 h to remove carbonates. The grain sizes of sediments (<4 μm (clay),

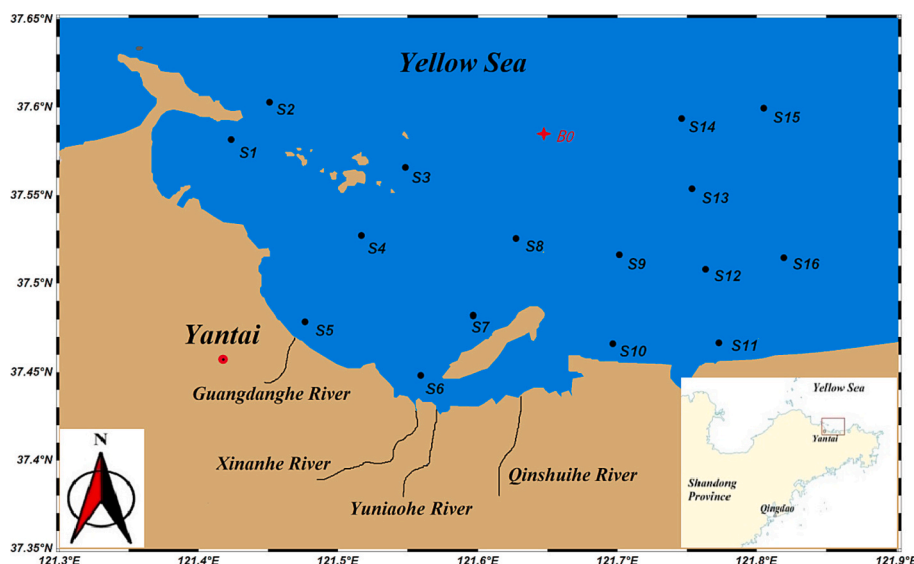


Fig. 1. Study area and sampling sites.

4–63 µm (silt), and >63 µm (sand)) were analyzed by a laser particle size analyzer (Mastersizer 2000F, Malvern, UK, RSD < 2.5 %) prior to treatment with 10 % H₂O₂ and 10 % HCl after freeze drying.

2.2.2. RIS fractionation extraction

The separation and determination of AVS, CRS, and ES were conducted following the modified cold diffusion procedure by Sheng et al. (2015). Briefly, AVS, CRS, and ES were separated sequentially by 9 mol/L HCl (1 h), acidic Cr(II) (2 h), and *N,N*-dimethylformamide + acidic Cr(II) + 9 mol/L HCl (1 h), respectively, under a pure N₂ atmosphere, at 60 ± 1 °C. The liberated H₂S was passively trapped in 0.1 mol/L CuCl₂ solution. The quantity of S for each solid-phase RIS species trapped in CuS was determined by iodometric titration. The precisions of the triplicate analyses were within 5 %.

2.2.3. Heavy metals analysis

SEM were measured by inductively coupled plasma mass spectrometry (ICP-MS, ELAN DRC II, PerkinElmer, Hong Kong, RSD < 0.08 %) after complete removal of H₂S in the above AVS test and filtration through a pre-rinsed 0.45 µm membrane. The geochemical fractions of heavy metals (Pb, Cu, Zn, and Cd) in samples were analyzed using modified Bureau Communautaire de Référence (BCR) sequential extraction procedure. The metal fraction was divided into the F1 (acid-soluble fraction), F2 (reducible fraction), F3 (oxidizable fraction) and F4 (residual fraction) by this method. F1, F2, F3, and F4 were separated sequentially by 0.11 mol/L HAc solution, 0.5 mol/L NH₂OH-HCl (pH = 2) solution, 30 % H₂O₂ (pH = 2–3) + 1 M NH₄Ac (pH = 2), and HNO₃ + HF + HClO₄ (7:4:1, v/v), respectively, and then analyzed by ICP-MS. Duplicate analysis on the extracts in sediments in Sishili Bay and its adjacent area (S4 and S12) was performed to control the quality of analytical data obtained from the BCR sequence extraction procedure. The RSD of heavy metals were <5 %.

2.3. Risk assessment methods of heavy metals

In order to assess the risks of heavy metals in sediments, the geoaccumulation index (*I*_{geo}) and risk assessment code (RAC) were employed. The relationship between the heavy metal level and the background level were judged by *I*_{geo} index, and the pollution degree of sediments by comparing the F1 and the total heavy metal content were assessed through the RAC values. The *I*_{geo} index and RAC divide the pollution degree into seven and five levels, respectively (Table S1). These indexes are calculated by the following equation:

$$I_{\text{geo}} = \log_2 [C_i / (1.5 \cdot B_i)]$$

$$\text{RAC} = C_m / C_i \times 100\%$$

where *C_i* is the concentration of a heavy metal, *C_m* is the F1 concentration of a heavy metal, *B_i* is the background value (here, the content in B0 was used), 1.5 is the background correction factor due to lithogenic effects (Kuang et al., 2022).

2.4. Statistical analysis

The figures were made by Origin Pro 2018 and Ocean Data View. Data normality was detected by the Kolmogorov-Smirnov (K-S) test. Principal component analysis (PCA) and Pearson correlation analysis were performed using Statistical Program for Social Sciences (version 26.0 for Windows). *P* < 0.05 and *P* < 0.01 were regarded as significant and extremely significant correlation level, respectively.

3. Results and discussion

3.1. Properties of sediments

The pH, EC, and salinity in mariculture sediments ranged from 8.45 to 8.64 (average 8.53 ± 0.05), 10.87 to 16.52 mS/cm (average 15.06 ± 1.48 mS/cm), and 6.0 to 9.0 ‰ (average 8.2 ± 0.7 ‰), respectively (Table S2). The TOC (average 0.42 ± 0.11 %) in sediments were close to the eastern Aegean coast mariculture areas in Turkey (Kucuksezgin et al., 2021), but much higher than the control point value (B0). This result indicated that mariculture activities would increase sediment organic carbon loading. Generally, approximately 40 % feed may enter the sediment in mariculture process, resulting in a huge accumulation of residual feed with high organic carbon in sediment. Meanwhile, some chemicals used in the breeding process and the excretion of breeding products can also lead to increases in TOC and nutrients in the sediments (Pan et al., 2019). Furthermore, organisms in the ocean can also transport TOC and nutrients to marine sediments through a series of natural processes such as growth, reproduction, death and decomposition (Meyers, 1997).

There was a significant linear correlation presented between TOC and TN (*r* = 0.894, *P* < 0.01), indicating that they were mainly derived from common organic source (Kucuksezgin et al., 2021). In order to further trace the source of organic matter, the ratio of TOC to TN (C/N) was used to differentiate the aquatic or terrestrial source. The values of C/N ranging from 5 to 7, and more than 12 indicate that the source of organic matter in sediment is marine planktonic organisms, and terrestrial input, respectively (Redfield et al., 1963; Meyers, 1997). The C/N in study area ranged from 7.67 to 14.00 (average, 10.29 ± 1.52), indicating that organic matter in sediments originated from terrestrial and marine sources (Kucuksezgin et al., 2021).

The grain sizes of sediments in study area are shown in Fig. 2. The average proportions of sediment fractions with clay, silt, and sand were 4.81 ± 1.47 %, 58.86 ± 14.97 %, and 36.33 ± 16.39 %, respectively. The grain sizes of sediments (except for S6) were dominated by sandy silt, and there was a coarse to fine transition from offshore region to deep water. This is likely a result of regional hydrodynamics (Hao et al., 2021). In the present case, this phenomenon can be explained by the combined influences of river input and coastal currents because there are 4 rivers along the coastal line in study area (Fig. 1). Therefore, the hydrodynamic condition in offshore region will be strongly influenced by tidal flows and coastal flows, resulting in the resuspension and

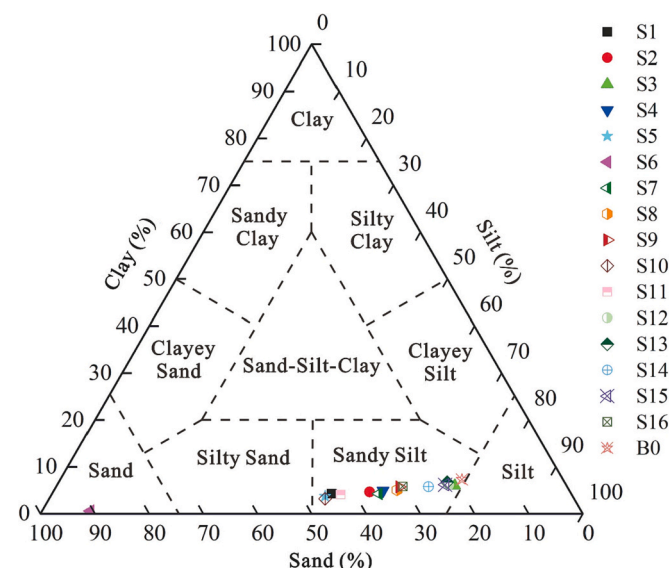


Fig. 2. Grain size of sediments in study area.

migration of fine-grained sediments, which may lead to the strong coarsening of sediments in offshore region (Wang et al., 2014). In contrast, S6 locates at an estuary of Xinnanhe River and has strong hydraulic disturbance, the sediment composition is mainly sand. In addition, compared with the sea cucumber farming area, the fine particle content in shellfish farming area is low, which is mainly related to the feeding mode of shellfish. Because shellfish belong to filter feeding organisms, they can ingest and retain fine particles with a size $>4\ \mu\text{m}$ (Ward and Shumway, 2004).

3.2. Distribution characteristics of RIS

The content of RIS in mariculture area ranged from 227.54 to 1397.08 mg/kg, with an average of $756.72 \pm 352.14\ \text{mg/kg}$ (Fig. 3), which was higher than that in control area (B0, 577.11 mg/kg). For RIS, AVS, CRS, and ES shared 31.54 % ($236.64 \pm 107.69\ \text{mg/kg}$), 46.40 % ($345.65 \pm 181.71\ \text{mg/kg}$) and 22.06 % ($174.44 \pm 96.65\ \text{mg/kg}$), respectively. Totally, the coefficient of variation (CV) of AVS (45.51 %), CRS (52.57 %), ES (55.41 %) all presented high values, indicating a great variation of RIS existed in sediments. Generally, accumulation of AVS in sediment is considered as a typical characteristic of sediment aging. Therefore, the aging degree of mariculture sediments would be closely related to AVS distribution. In this work, the AVS in study area was close to Sanggou Bay for scallop mariculture, Zhangzi Island for scallop and sea cucumber mariculture (Kang et al., 2016), but significantly higher than the site B0. This result could be explained that mariculture process might lead to the accumulation of organic matter in sediments, which would stimulate benthic mineralization and oxygen consumption, and consequently promotes dissimilatory sulfate reduction and AVS production (Dahlbäck and Gunnarsson, 1981). The lowest AVS was recorded at S6 because there was low TOC content ($0.11 \pm 0.01\%$) (Table S2), large sediment particle size (sand $>90\%$), and strong hydraulic disturbance near the estuary and shore, which was not conducive to the formation of AVS.

The determination method of AVS in this study is an optimization of

the iodometric method used for sulfide determination in State Marine Sediment Quality Standard in China (GB 18668-2002), and the optimized AVS determination is faster and more accurate than conventional iodometric method. Therefore, the AVS values obtained in this study can be evaluated using corresponding levels in GB 18668-2002. Totally, AVS values in 35 % sampling sites were inferior to the first level of sulfide in GB 18668-2002, indicating that the sediments had been polluted by AVS. Compared to the sea cucumber farming area, the AVS in shellfish farming area presented high values, especially in S1, S2, S3, S4, and S7. This result indicated that different culture types would result in different sediment aging processes. Biological sedimentation in shellfish farming activities can cause rapid deposition of particulate matter. In Sishili Bay, the deposition rate of particulate matter in scallop farming was 2.46 times higher than natural sedimentation (Zhou et al., 2006), and high deposition rates would favor the retention of AVS in sediments (Gagnon et al., 1995). In contrast, deposit-feeding sea cucumbers not only consumed large amounts of organic matter in the sediment (Purcell et al., 2016), but also disturbed and translated the sediment during the feeding process (Roberts et al., 2000), which was unfavorable to the generation and retention of AVS. Therefore, the AVS in shellfish farming area was higher than that in sea cucumber area.

Overall, the CRS dominated the RIS, which probably could be explained that there was a high transformation degree from AVS to CRS existed in sediments. To verify this deduction, the value of AVS/CRS ratios, a widely used indicator for assessing the conversion of AVS to CRS was calculated (Gagnon et al., 1995). The AVS/CRS ratios in study area were lower than 1 (average 0.7), indicating high conversion from AVS to CRS occurred. Moreover, the low ES (accounting for 21.34 % of RIS) also indicated that the AVS could be transferred to CRS with ES through rapid geochemical conversion (Sheng et al., 2015). As a product of incomplete oxidation of sulfide by oxidizer (e.g. O_2 , NO_3^- , Fe^{3+}), ES is the most abundant short-term sulfate reduction product in sediments (Sheng et al., 2015), it can react with AVS to form CRS.

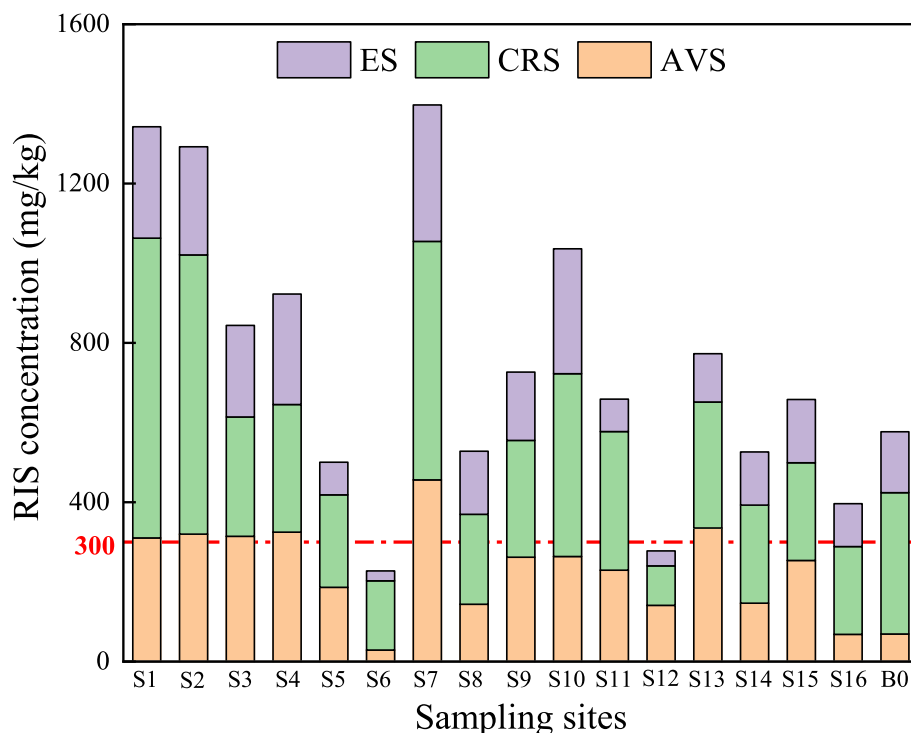


Fig. 3. Distribution of different fractions of RIS in sediments (red line: the first class standard value of AVS in GB 18668-2002). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3.3. Distribution and ecological evaluation of heavy metals

The average values of total heavy metals in sediments in study area decreased according the following sequence: Zn (82.55 ± 13.90 mg/kg) > Pb (25.44 ± 1.75 mg/kg) > Cu (19.70 ± 3.52 mg/kg) > Cd (0.42 ± 0.10 mg/kg) (Table S3), which were superior to the corresponding standard levels of heavy metals in GB 18668-2002. The CV values of Pb, Cu, Zn, and Cd were low, indicating these metals exhibited a homogeneous distribution in study area. PCA and correlation have often been performed to identify potential origins of multiple elements in sediments. Three principal components were identified with eigenvalues > 1, accounting for 89.605 % of the total variance, indicating good performance in PCA (Table S4). PC1 was mainly loaded with TOC, fine particles (silt + clay), and Cu, explaining 46.095 % of the total variance. PC1, TOC, fine particles, and Cu exhibited high loads of 0.852, 0.874, and 0.930, respectively. Furthermore, the correlation between any two elements of them was significant ($P < 0.01$) (Table S5). The results in this work were consistent with previous reports, mariculture activities resulted in the increase of TOC and the coarsening of grain sizes in sediments (Chen et al., 2016). Moreover, most bait and feed used in aquaculture often contain Cu, and it is also related to disinfectant usage in aquaculture process (Fan et al., 2022). Accordingly, PC1 always represent mariculture activities. In PC1, Zn also had high loads (0.578), indicating that Zn originated partly from mariculture activities. Feed used in aquaculture also usually contain Zn (Fan et al., 2022). PC2 was characterized by Zn and Cd, with the significantly correlative ($P < 0.05$) (Table S5). In PC2, Zn and Cd exhibited high loads of 0.664 and 0.949, respectively. In this study, the areas with high Cd contents are mostly located near the channel. There are many ships often use Zn-Al-Cd alloy sacrificial anodes to prevent corrosion. Thus, PC2 represents pollution from ship transportation. The Pb was the only element to score high on

PC3. As analysis of Pearson correlation and PCA displayed, Pb did not show a significant relationship with any heavy metals, TOC, and fine particles (Table S5). Furthermore, there was no obvious terrestrial or artificial source could be identified, so the Pb could probably originated from atmospheric deposition, because Pb had been widely considered to be an important source of Pb in marine sediments (Pan et al., 2020). Thus, PC3 represents pollution from atmospheric deposition.

For the geochemical fractions, the F4 dominated total heavy metals with mean values of 75.85 %, 68.94 % and 63.71 % of total Zn, Cd and Cu, respectively (Fig. 4). The high F4 indicated that they posed minor potential threat to marine biota. However, the dominated fraction of Pb was F2 (47.38 %), followed by F4 (40.80 %), F3 (6.37 %), and F1 (5.45 %). This result was similar to that in Zhelin Bay mariculture areas because of the strong adsorption of iron manganese oxide on Pb (Gu and Lin, 2016). Previous studies reported that Pb was adsorbed more strongly than any of the other ions ($Pb^{2+} > Cu^{2+} > Zn^{2+} > Cd^{2+}$) by iron manganese oxide (McKenzie, 1980). Meanwhile, the combination of iron manganese oxide and heavy metals will be influenced by pH (Du Laing et al., 2009). In study area, the pH of sediment is alkaline (mean value 8.53), which favors the adsorption of Pb by iron manganese oxide. For Cd, F1 was the second abundant form (21.47 %), which could be related to the high concentration of HCO_3^- in sediments (Gu, 2018). The second high percentages of Cu (22.72 %) and Zn (9.81 %) were found in F2. And all heavy metals accounted for the low proportions (<12 %) in F3 at all sites. Overall, the average proportions of biological effective forms (F1 + F2 + F3) of heavy metals in sediments decreased according the following sequence: Pb (59.20 %) > Cu (36.76 %) > Cd (31.06 %) > Zn (24.15 %).

As shown in Fig. 5, the average I_{geo} index for Pb (−0.61), Cu (−0.73), Zn (−0.78), and Cd (−0.51) were < 0. Based on the categories (Table S1), the sediments in study area were at unpolluted level ($I_{geo} \leq$

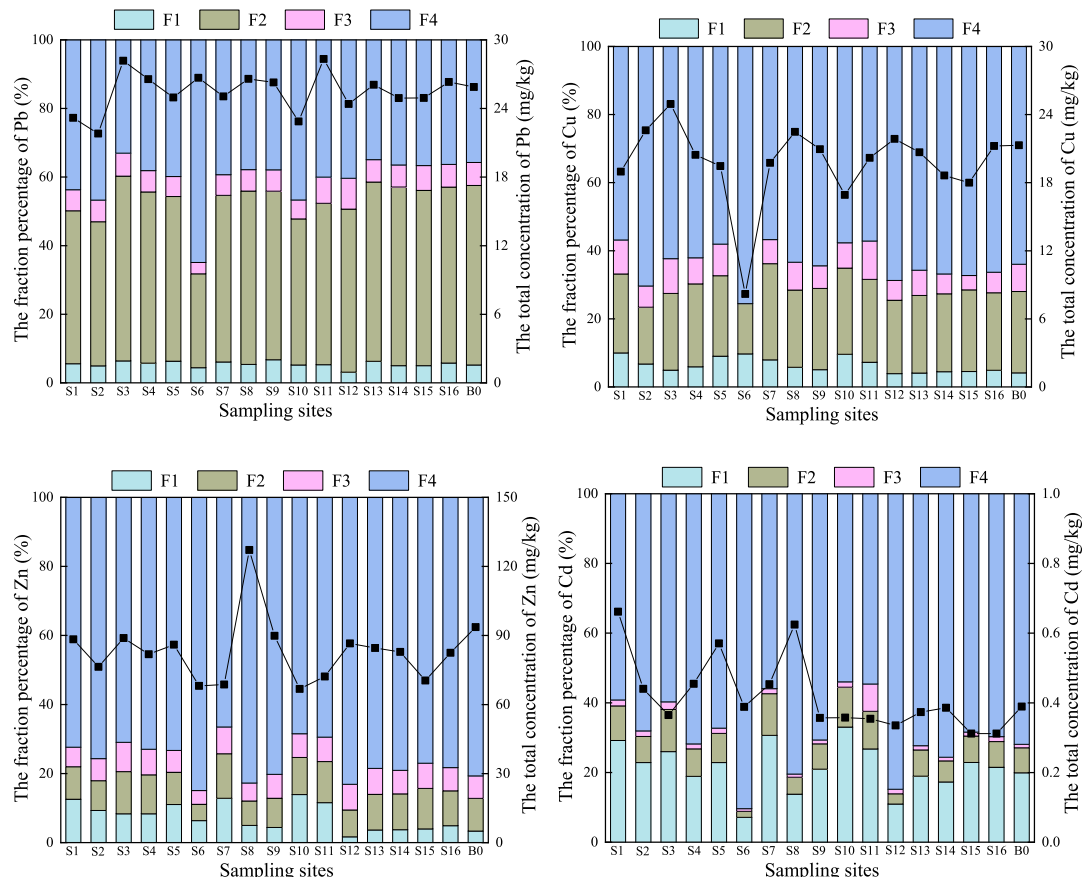


Fig. 4. Distributions of total heavy metals and different geochemical phases in sediments.

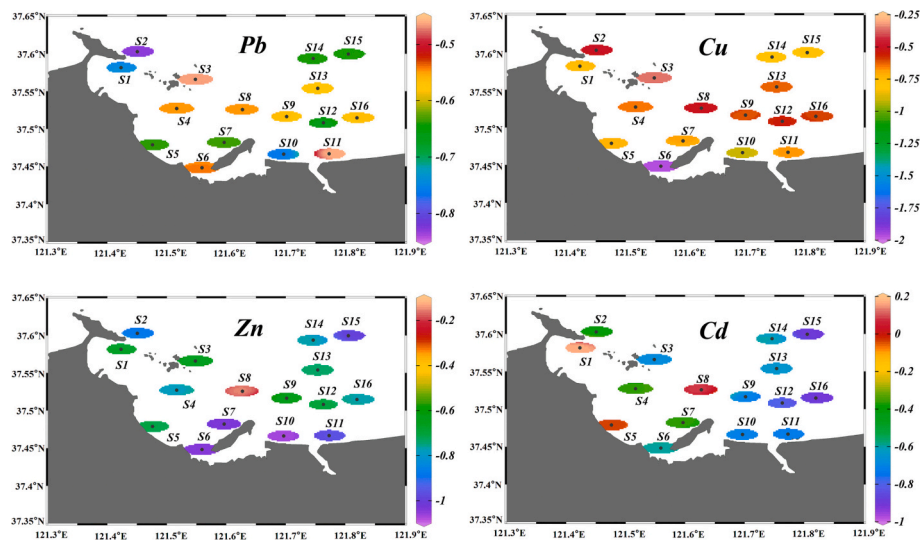


Fig. 5. The I_{geo} values of heavy metals in sediments in study area.

0). Besides their total content, the pollution of heavy metals depends on their specific chemical forms (Liu et al., 2022). The RAC, a widely used and effective indicator based on the content of a substance in the acid-soluble fraction, was used. In contrast to the result of I_{geo} index, the average RAC of Cd (21.48 %) was higher than 10 %, presenting a medium risk (Fig. 6). The RAC of Pb (5.45 %), Cu (6.49 %), and Zn (7.60 %) were below 10 %, they posed low risks to the environments. Although the proportions of F1 of Pb, Cu, and Zn were high, the risks were almost negligible due to their low total amounts.

Due to the close relationship between AVS and heavy metals, SEM/AVS is also often used to predict the toxicity of heavy metals in the mariculture sediments. In this work, the ratios of SEM/AVS were <0.5 at all sampling sites, indicating there was no adverse effect by heavy metals. Interestingly, the concentrations of SEM_{Pb} , SEM_{Cu} , and SEM_{Cd} were approximately consistent with the sums of F1, F2, and F3 of Pb, Cu, and Cd. However, the sums of F1, F2, and F3 of Zn were mostly lower than the concentrations of SEM_{Zn} . This may be due to a part of the stable residual fraction of Zn has been leached during AVS extraction (Xia

et al., 2016). Since the residual fraction is fairly stable and unlikely to be released into environment (Liu et al., 2022), so the toxicity of heavy metals may have been overestimated, especially for Zn. Therefore, the ratio of heavy metals biological effective forms (F1 + F2 + F3) to AVS ($\sum(F1 + F2 + F3)/AVS$) should be suggested to assess the toxicity of heavy metals in the mariculture sediments. The biological effective forms can not only evaluate the biotoxicity of heavy metals individually, but also as a sum provide a general index to judge the toxicity of heavy metals in the mariculture sediments (Xia et al., 2016).

3.4. Correlation analysis for RIS and heavy metals

The Pearson correlation analysis (Fig. 7) showed that AVS/RIS was positively correlated with the proportion of F2 of Pb ($F2/(F1 + F2 + F3 + F4)$, $F2/\sum F$) ($P < 0.05$) and Pb ($F3/\sum F$) ($P < 0.01$) in sediments, indicating that the accumulation of AVS was conducive to the F2 and F3 formations of Pb. Interestingly, there was a significant negative correlation ($P < 0.01$) between AVS/RIS and Pb ($F4/\sum F$) ($P < 0.05$). This

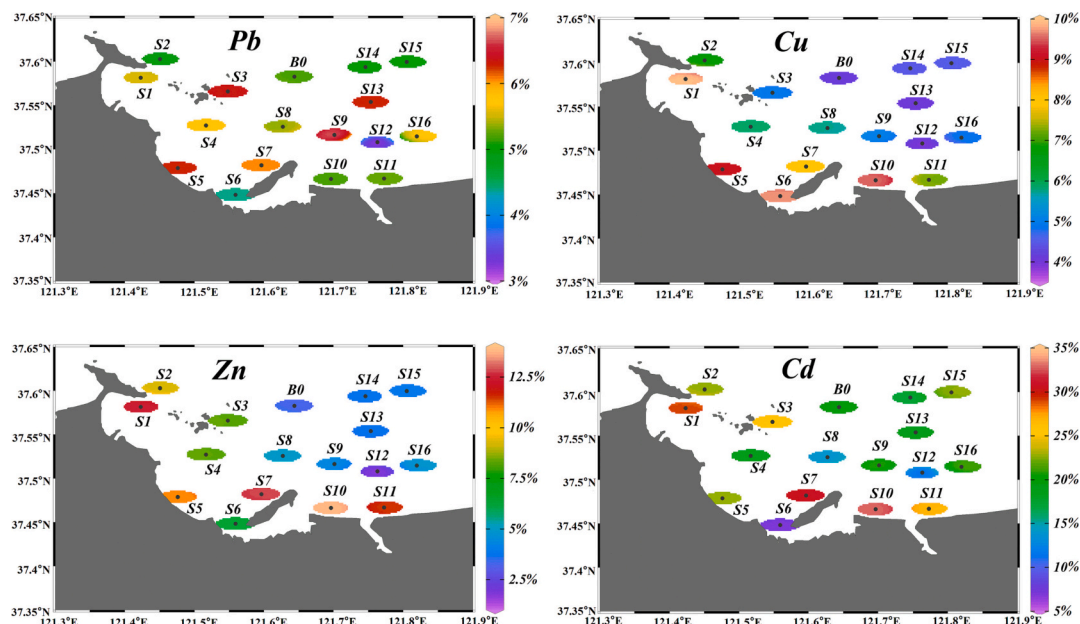


Fig. 6. The RAC values of heavy metals in sediments in study area.

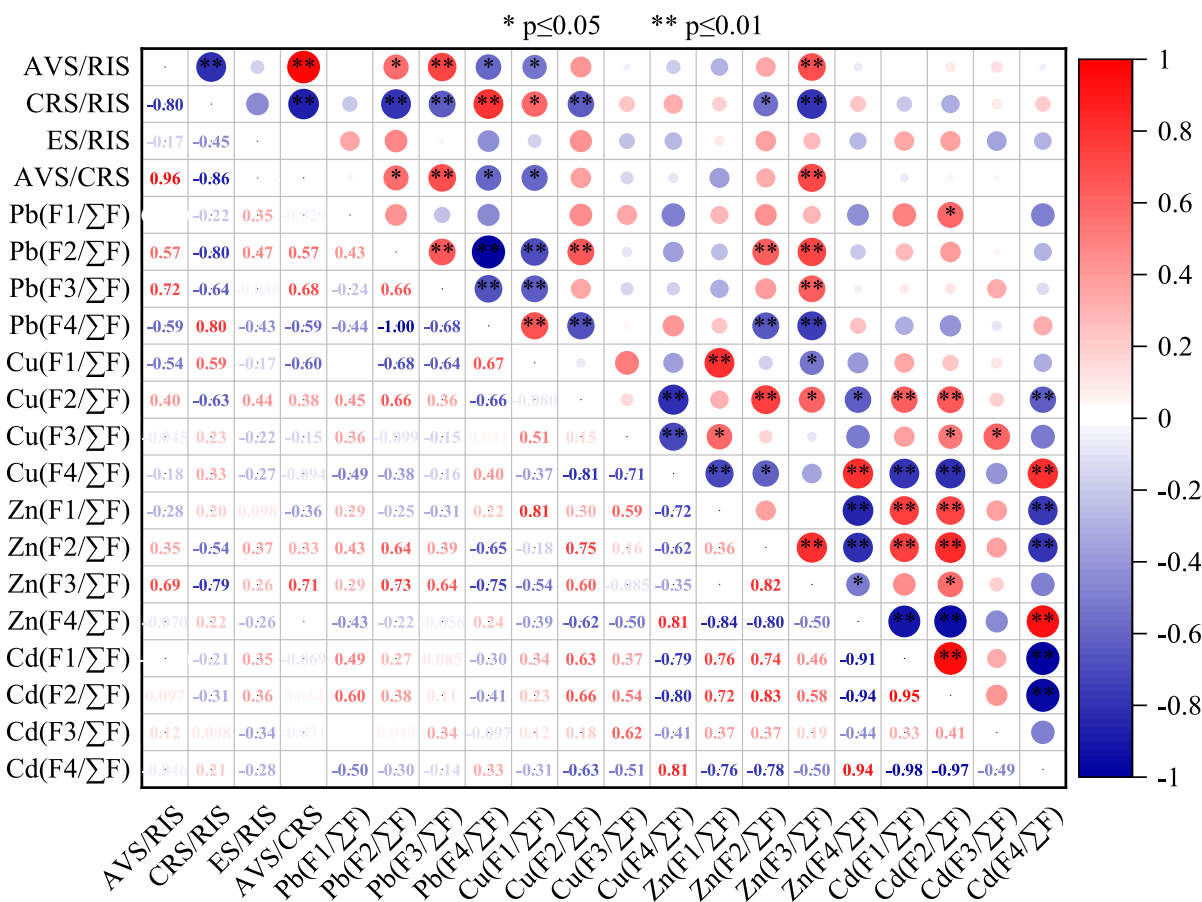


Fig. 7. Correlation analysis for the RIS and the geochemical fractions of heavy metals in sediments in study area.

may be attributed to different pollution degrees and TOC loads. Previous study demonstrated that predominance of F4 for Pb was mainly encountered in non-polluted sediments, while in contaminated sediments, F4 of Pb was inversely proportional to the degree of contamination (Woszczyk and Spychalski, 2013). In this work, low F4 showed that mariculture activities would cause heavy metal pollution to sediments. Additionally, TOC was significantly negatively correlated with F4 of Pb ($r = -0.6801$, $P < 0.01$), while high TOC content was favorable to the generation and retention of AVS. There was a significant negative relationship presented between AVS and the F1 of Cu ($P < 0.05$) in sediments, indicating the accumulation of AVS in mariculture sediments promoted the conversion of F1 of Cu to other fraction and improved the stability of Cu. The solubility of heavy metals decreased according the following sequence: ZnS ($K_{sp} = 2.0 \times 10^{-22}$) > CdS ($K_{sp} = 8.0 \times 10^{-27}$) > PbS ($K_{sp} = 8.0 \times 10^{-28}$) > CuS ($K_{sp} = 6.3 \times 10^{-36}$) (Fang et al., 2018). Meanwhile, the stability of F1 is the worst compared to other fractions, so AVS prefers to react with F1 of Cu. AVS has strong correlation with F3 of Pb and Zn, but weak correlation with F3 of Cu. This may be due to F3 of Pb and Zn was dominated by sulfide bound states (Chen et al., 2020), while the F3 of Cu may be dominated by organic matter bound states. Cu has the highest complexation stability constant with organic matter ($\lg K_{\text{Cu}} > \lg K_{\text{Pb}} > \lg K_{\text{Zn}} > \lg K_{\text{Cd}}$) and it is easy to be complexed (Dinu, 2013). Meanwhile, mariculture activities often lead to the accumulation of TOC in sediment. So the significant correlation ($r = -0.7073$, $P < 0.01$) between TOC and F3 of Cu confirmed the above hypothesis.

The correlations of CRS/RIS and AVS/CRS with the proportion of each heavy metal fraction showed that the conversion process of AVS to CRS promoted the conversion of F2 and F3 to F4 for Pb, F2 to F1 for Cu, and F2 and F3 to other fraction for Zn. These phenomena indicated that increased Pb stability and decreased Cu stability led to the reduction of mariculture sediments aging process. And the decreases of F2 of Cu, Pb

and Zn were mainly related to the formation mode of CRS in sediments. Generally, transforms from AVS to CRS primarily occur through the polysulfide pathway ($\text{FeS} + \text{S}^0 = \text{FeS}_2$) and H_2S pathway ($\text{FeS} + \text{H}_2\text{S} = \text{FeS}_2 + \text{H}_2$) (Jiang et al., 2021b). Both methods require the participation of ES, which is often produced by incomplete oxidation of RIS by oxidants such as oxygen and iron manganese oxide (Sheng et al., 2015). In this work, for above process, iron manganese oxides could be consumed during the incomplete oxidation of RIS, resulting in the release of F2 of heavy metals absorbed by them. The decreases of F3 for Pb and Zn could be due to some PbS and ZnS were trapped in the residual phases during the conversion of AVS to CRS.

4. Conclusion

Organic matter in mariculture area originated from terrestrial and marine sources. The sediment grain sizes mostly dominated by silt. Mariculture activities enhanced AVS accumulation, and shellfish farming areas presented higher values than sea cucumber. Based on PCA and correlation analysis, three PCs were identified to represent different heavy metals sources: mariculture activities (Cu and Zn), ship transportation (Cd and Zn), and atmospheric deposition (Pb). The environmental risk of heavy metals in mariculture area was low. The sums of F1, F2, and F3 of Zn were mostly lower than the SEM_{Zn} . Since SEM/AVS would overestimate the toxicity of heavy metals, the ratio of $(\text{F1} + \text{F2} + \text{F3})/\text{AVS}$ should be selected to assess the toxicity of heavy metals in mariculture sediments. The AVS accumulation was conducive to the formation of F2 and F3 of Pb, F3 of Zn, while unfavorable to F4 of Pb, and F1 of Cu. Such inconsistency was also observed in the CRS accumulation. Increased Pb stability and decreased Cu stability led to the reduction of mariculture sediments aging process. This study could provide theoretical supports for evaluating the process of mariculture

sediment aging, environment risk assessment and pollutant behavior identification, especially for sulfur and heavy metals. Furthermore the results could be used as valuable reference for the formulation of mariculture development and management policy normalization.

CRedit authorship contribution statement

Zhaoran Li: Writing – original draft, Investigation, Data curation. **Tao Ma:** Investigation, Writing – review & editing. **Yanqing Sheng:** Writing – review & editing, Conceptualization, Funding acquisition, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

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